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Research paper

Experimental evaluation of cognitive agents for collaboration in human-autonomy cyber defense teams

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ABSTRACT

Autonomous agents are becoming increasingly prevalent and capable of collaborating with humans on interdependent tasks as teammates. There is increasing recognition that human-like agents might be natural human collaborators. However, there has been limited work on designing agents according to the principles of human cognition or in empirically testing their teamwork effectiveness. In this study, we introduce the Team Defense Game (TDG), a novel experimental platform for investigating human-autonomy teaming in cyber defense scenarios. We design an agent that relies on episodic memory to determine its actions (*Cognitive agent*) and compare its effectiveness with two types of autonomous agents: one that relies on heuristic reasoning (*Heuristic agent*) and one that behaves randomly (*Random agent*). These agents are compared in a human-autonomy team (HAT) performing a cyber-protection task in the TDG. We systematically evaluate how autonomous teammates' abilities and competence impact the team's interaction and outcomes. The results revealed that teams with Cognitive agents are the most effective partners, followed by teams with Heuristic and Random agents. Evaluation of collaborative team process metrics suggests that the cognitive agent is more adaptive to individual play styles of human teammates, but it is also inconsistent and less predictable than the Heuristic agent. Competent agents (Cognitive and Heuristic agents) require less human effort but might cause over-reliance. A post-experiment questionnaire showed that competent agents are rated more trustworthy and cooperative than Random agents. We also found that human participants' subjective ratings correlate with their team performance, and humans tend to take the credit or responsibility for the team. Our work advances HAT research by providing empirical evidence of how the design of different autonomous agents (cognitive, heuristic, and random) affect team performance and dynamics in cybersecurity contexts. We propose that autonomous agents for HATs should possess both competence and human-like cognition while also ensuring predictable behavior or clear explanations to maintain human trust. Additionally, they should proactively seek human input to enhance teamwork effectiveness.

1. Introduction

With advances in computational power, network robustness, cognitive modeling, and machine learning capabilities, a new form of team is on the rise: the human-autonomy team (HAT; McNeese, et al., 2018). HATs are composed of at least one member of a team that meets the definition of an 'autonomous agent', another member is human, and the team members depend on each other to achieve a collective goal. Wynne and Lyons (2018) define an 'ideal' autonomous agent teammate as: 'a highly altruistic, benevolent, interdependent, emotive, communicative and synchronized agent teammate, rather than simply an instrumental tool'. Although existing work has examined the effects of agent performance (Bansal et al., 2019) and warmth (Harris-Watson, Larson, Lauharatanahirun, DeChurch, & Contractor, 2023) on the effectiveness of HAT, there is increasing recognition that human-likeness offers unique ways to improve human attitude toward agents and facilitates human–agent cooperation (Glikson & Woolley, 2020; Pelau, Dabija, & Ene, 2021; Zhang, Chong, Kotovsky, & Cagan, 2023). Human participants have shown a higher level of trust in the aid of computer agents with a human-like appearance (de Visser et al., 2012; Von der Pütten, Krämer, Gratch, & Kang, 2010), verbal communication (Kulms & Kopp, 2019), and a display of emotion (Boone & Buck, 2003; Kay, Keller, & Lehmann, 2020). However, there is very limited information, empirical investigations and data regarding how

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humans and autonomous agents that are human-like at the *cognitive level* collaborate (Gonzalez, 2024). Musick, O'Neill, Schelble, McNeese, and Henke (2021) explored how team composition affects team processes and emergent team cognition. However, their teams consisted of only humans, and used a Wizard of Oz approach to make participants believe their teammates were autonomous agents. In contrast, our study implements genuine human-autonomy teams with actual autonomous agents working alongside human participants, providing empirical evidence about the true human-autonomy interaction rather than merely perceived interaction.

Cognitive architectures and theories of human decision making have made significant progress in emulating human-like behavior in dynamic environments (Ritter, Tehranchi, & Oury, 2019). Unlike typical computational algorithms that aim to make optimal decisions, cognitive architectures adhere to human constraints such as forgetting, limited attention, and bounded rationality (Gonzalez, 2024). Cognitive models such as instance-based learning theory (IBLT; Gonzalez, Lerch, & Lebiere, 2003) have been implemented in various domains and demonstrated similarity to human decision making processes, including repeated binary choice tasks (Gonzalez & Dutt, 2011; Lejarraga, Dutt, & Gonzalez, 2012), sequential decision-making (Bugbee & Gonzalez, 2022), and practical applications such as identifying phishing emails (Cranford, Lebiere, Rajivan, Aggarwal, & Gonzalez, 2019) and making decisions about cyber attacks (Aggarwal et al., 2022). More recently, cognitive agents have demonstrated a human-like theory of mind (Nguyen & Gonzalez, 2022), or the natural ability to predict the intentions and false beliefs of other agents (Geib & Goldman, 2009; Kautz, Allen et al., 1986). ToM has also been shown to be essential for HAT teamwork (Bendell, Williams, Fiore, & Jentsch, 2024)

The questions we pursue in this research are as follows: Do human-like cognitive agents have any advantage over optimally performing non-cognitive agents in HAT collaborations? How do humans perceive the cooperativeness and trustworthiness of cognitive and non-cognitive agents in HATs? We explore the potential of cognitive theory to build human-like agents for HATs and compare such cognitive agents in a collaborative HAT applied to cyber defense. We developed a cognitive model that represents the human decision process and incorporated this agent into a HAT experiment where humans and autonomous agents interact and collaborate as a team to ensure the security of an computer network. To enable this investigation, we designed the Team Defense Game (TDG), a novel experimental platform that facilitates controlled studies of human-autonomy teaming in cyber defense scenarios. In an online experiment, we compare how a human defends against adversaries with the help of an autonomous defender that acts randomly, uses smart heuristics, or learns from experience interactively in the task (i.e., the cognitive agent). To evaluate team effectiveness (Hackman, 1978; O'Neill, Flathmann, McNeese, & Salas, 2023), we measure the HATs' performance in terms of their ability in a cyberdefense scenario that requires agents to prevent attacks and resolve network issues. In addition to performance in this task, we also measure human perception of agents in terms of trustworthiness and cooperativeness in a post-experiment questionnaire (Kocielnik, Amershi, & Bennett, 2019; Ragot, Martin, & Cojean, 2020). To better understand how different types of autonomous agents lead to different team outcomes, we measure three collaborative metrics during the teamwork process based on the dynamics of the cyberdefense task. At the general level, we hypothesize that the cognitive agent will be the best teammate, capable of cooperative interaction with humans during teamwork, lead to the best team performance, and be perceived as the most trustworthy and cooperative partner by humans. The following review of related work supports this general expectation.

2. Related work

2.1. Human-Autonomy Teaming (HAT) in cybersecurity

Autonomous systems have been used to control cyber operations (Stevens, 2020) and network security challenges (Bécue, Praça, & Gama, 2021). Deep learning techniques have been used for the detection of anomalies and malware (Tayyab, Khan, Durad, Khan, & Lee, 2022). Bayesian networks have been applied for the identification of attack paths and the correlation of incident intrusion (Albasheer et al., 2022). Game-theoretic methods have been used to model the interaction between the defender and the adversary as security games and offer optimal allocation strategies of defense resources (Fang, 2021). However, it is clear that across cyber defense, many automated components are limited forms of adaptable automation; that is, they have low adaptability, or the adaptability is overly time-consuming. A typical example of the latter is an intrusion detection system in which a user configures alerts but has to manually adjust and maintain those settings and manage potential false alarms. Recently, the application of reinforcement learning has allowed adaptive cyber defense that is flexible to the dynamics of network/system security status (Du, Song, Milani, Gonzales, & Fang, 2022). However, much of this work still uses autonomous systems as decision-support tools. In this type of work, the autonomous system has no agency and is used to give recommendations to humans rather than to work with humans in a HAT collaboration.

Today, cyber analysts are a scarce resource and are often overloaded (Nobles, 2022). Security Operations Centers (SOCs) combat the growing problem of alert fatigue, where the sheer volume of alerts overwhelms SOC analysts and raises the risk of overlooking critical threats (Chhetri et al., 2024), creating ideal conditions for misallocation of attention (Parasuraman, Molloy, & Singh, 1993). To address this challenge and meet the demands posed by sophisticated adversaries and the need for agile responses, autonomous systems must evolve beyond mere recommender systems and operate with higher levels of agency (Linkov et al., 2023). The cyberdefense technology community is beginning to recognize the necessity of building autonomous agents that can act on their own (Kott, 2023).

However, it is essential to explore autonomous agents that can account for the decision-maker's values or specific mission needs. For example, following a cyber attack, an AI-generated decision engine may recommend disabling an application on the compromised computer system. This action may neutralize the threat posed by the compromised system, but could simultaneously endanger a mission, negatively impact a user's ability to perform critical tasks, or allow the adversary to extend the duration or scope of the cyber attack (Linkov et al., 2023). Human experts should stay in the loop to provide intuition, critical thinking, and contextual information by approving or denying recommendations from AI decision engines that may have negative impacts. Sarker, Janicke, Mohammad, Watters, and Nepal (2023).

In summary, autonomous agents should be able to operate with a degree of self-autonomy and self-directed behavior (agency) while at the same time working interdependently with humans to achieve a shared objective. For the partnership to be successful, the potential benefits of HAT must be weighed against foreseeable negative human-autonomy interactions. Unintended results of incorporating autonomous agents that must be addressed include creating more (not less) work for humans, failing to decrease required manpower, deskilling operators, reducing awareness, contributing to accidents, and loss of life (Gutzwiller, Clegg, & Blitch, 2013; Lyn Paul, Blaha, Fallon, Gonzalez, & Gutzwiller, 2019; Strauch, 2017). The critical factors for successful teamwork in HAT applications must be identified.

The growing literature on HATs in domains such as urban search and rescue (Wohleber, Stowers, Barnes, & Chen, 2023) and hospital management (Chiou, Lee, & Su, 2019) has identified some of these critical factors for successful teamwork, but little is known about HATs in cybersecurity. Teams in cybersecurity operations, especially those in 24/7 security operations centers, have specific dynamics (Paul, 2014). Different cyberdefense scenarios also pose unique challenges for humans and autonomous agents working in teams. For example, in incident response and recovery, autonomous agents might focus on information triage. They would leave the task of further analysis and strategic decision-making to humans. In adaptive defense, autonomous

agents can be more efficient in dynamically adjusting security mechanisms based on real-time threat intelligence. Humans would supervise and fine-tune agent decisions only when necessary (Linkov et al., 2023). Due to these varied and unique applications, a synthetic cyber task environment is needed to empirically evaluate HATs with different team compositions in various cyber scenarios. This environment would also provide humans with early exposure to demonstrations of simulated HATs before implementing them in the cyber workforce.

2.2. Cognitive models: Computational representation of human decision processes

The ability of autonomous systems to emulate human decision-making can benefit human-autonomy teaming (Jiao, Zhou, Gebraeel, & Duffy, 2020; McNeese, Demir, et al., 2018; Prebot, 2020). As suggested by Gutzwiller, Espinosa, Kenny, and Lange (2018) and Zhang, McNeese, Freeman, and Musick (2021), creating autonomous agents that can work efficiently with humans should involve modeling team interaction and human cognition. It has the potential to ease the coordination of actions, improve trust in autonomous agents, and increase the performance of the team. In fact, models of human cognition have already been used in tutoring systems, playing the role of a "simulated student". Computer tutors using cognitive models of students to build teaching instructions and provide directed feedback were shown to improve student performance by the same amount as conventional methods one third of the time (Anderson, Corbett, Koedinger, & Pelletier, 1995; Ferster, 2022; Matsuda et al., 2013).

Existing methods in human-autonomy teaming typically do not involve modeling the cognitive mechanisms that underlie dynamic decision making (Ren, Chen, & Qiu, 2023). As a result, the actions of these autonomous teammates can be difficult for end users to understand, even if they are theoretically more optimal than the decisions made by more cognitively inspired agents (Li et al., 2023). Overall, the research on decision support systems in cyber defense has put a strong preference for optimal decisions, rather than understandable and human-like decision making (Vegesna, 2023). In this work we investigate the ability of cognitively inspired autonomous agents to integrate with humans into a team, and compare it to more a optimally designed Heuristic model, which is described more fully in the following sections. Another motivation supporting the improved teaming afforded by more cognitively inspired agents, compared to deterministically optimal ones, is that they can better adapt to the natural variation in human behavior and theoretically result in improved teaming performance.

Instance-Based Learning Theory (IBLT) emerged from the need to explain dynamic human decision-making processes, where a sequence of interdependent decisions is made sequentially (Gonzalez et al., 2003). IBLT provides a general algorithm and mathematical formulation of memory retrieval related to the well-known cognitive architecture ACT-R (Anderson & Lebiere, 2014). The theory proposes a representation of decisions and outcomes in the form of instances. In the past decade, cognitive models based on IBLT have been applied to represent the dynamic decision-making process in various domains that require real-time interactivity between models and humans (Nguyen, Phan, & Gonzalez, 2023). With this increased use of IBLT, the application of models to tasks that involve multiple players is also becoming more common. The initial theoretical developments of IBLT in this direction involved two-person game theoretical models (Gonzalez, Ben-Asher, Martin, & Dutt, 2015). More recently, other interesting applications of IBLT models have been proposed, including the ability to predict other agents' goals, beliefs, and intentions through the Theory of Mind reasoning (Nguyen & Gonzalez, 2020). In summary, existing work on learning-based cognitive modeling, specifically IBLT, has shown that these models operate in ways similar to humans. This similarity can enhance trust and understandability in HATs, as it allows more relatable and predictable interactions between human operators and autonomous agents. In addition, researchers can use the same mechanisms to build human-like models of the theory of mind by observing others' behavior. Thus, IBLT provides critical building blocks for modeling shared cognition processes—memory, attention, and reasoning—central to Human-Autonomy Teaming.

In the context of cyber security, IBL models have been developed to represent individual human cyber defense decisions (Dutt, Ahn, & Gonzalez, 2011), human attacker decisions that can inform cyber defense strategies (Cranford, Gonzalez, et al., 2020; Cranford, Gonzalez, Aggarwal, et al., 2020; Gonzalez, Aggarwal, Lebiere, & Cranford, 2020), and end-user phishing classification decisions that can help improve cyber defense (Cranford et al., 2019; Xu, Singh, & Rajivan, 2022). However, there is no existing work that incorporates IBLT for the defense of human-autonomy teams.

2.2.1. Instance-based learning theory

Although both the process and the mechanisms of IBLT have been published in multiple papers, we reproduce the mathematical formulations of the theory here for completeness. The central element of IBLT is the "instance". It represents a unit of memory resulting from evaluating potential choice alternatives. Each decision is stored in an instance, structured with three elements that are built over time: a situation state s which is composed of a set of features f; a decision or action a taken corresponding to an alternative in state s; and an expected utility or experienced outcome x of the action taken in a state. Concretely, for an IBL agent, an option $k = (s, a)$ is defined by action a in state s. At time t, assume that n_{kt} different instances (k_i, x_{ik_it}) for $i = 1, \ldots, n_{kt}$, associated with k. Each instance i in memory has an *Activation* value, which represents the ease of retrieving this information from memory (Anderson & Lebiere, 1998). Here, we consider a simplified version of the Activation equation, which only captures recency, frequency, and noise in memory:

$$\Lambda_{ik_it} = \ln\left(\sum_{t' \in T_{ik_it}} (t - t')^{-d}\right) + \sigma \ln \frac{1 - \xi_{ik_it}}{\xi_{ik_it}}, \tag{1}$$

where d and σ are the decay and noise parameters, respectively, and $T_{ik_it} \subset \{0, \ldots, t-1\}$ is the set of previous timestamps in which instance i was observed. The rightmost term represents noise to capture individual variation in activation, and ξ_{ik_it} is a random number drawn from a uniform distribution $U(0, 1)$ at each step and for each instance and option.

Activation of an instance i is used to determine the probability of retrieving an instance from memory. The probability of an instance where a soft-max function defines i:

$$P_{ik_it} = \frac{e^{\Lambda_{ik_it}/\tau}}{\sum_{j=1}^{n_{kt}} e^{\Lambda_{jk_jt}/\tau}}, \tag{2}$$

where τ is the Boltzmann constant (i.e., the "temperature") in the Boltzmann distribution. For simplicity, τ is often defined as a function of the same σ used in the activation equation $\tau = \sigma\sqrt{2}$.

The expected utility of option k is calculated based on *Blending* as specified in discrete choice tasks (Gonzalez & Dutt, 2011):

$$V_{kt} = \sum_{i=1}^{n_{kt}} P_{ik_it} x_{ik_it}. \tag{3}$$

The choice rule is to select the option corresponding to the maximum blended value. When the agent receives delayed results, the agent updates the expected utilities using a credit assignment mechanism (Nguyen, McDonald, & Gonzalez, 2021).

The Instance-Based Learning process is formalized in Algorithm 1, which outlines the sequential decision-making procedure of an IBL agent. The algorithm begins with initialization parameters that include a default utility, an empty memory dictionary to store instances, counters to track time and steps, and a flag indicating whether feedback is delayed. For each decision cycle, the agent observes the current state and enters an execution loop where it explores the available options.

Within this exploration, the agent calculates the activation values for each instance using Eq. (1), computes the retrieval probabilities with Eq. (2), and determines the blended values through Eq. (3). The agent then selects the action corresponding to the maximum blended value, executes it, observes the resulting state, and receives feedback. This feedback is stored in memory as a new instance, and the outcomes are updated through a credit assignment mechanism when the feedback is delayed. The process continues until a terminal state is reached or the step limit is exceeded, embodying the core principles of experience-based learning through memory activation and utility maximization.

Input: default utility u_0, a memory dictionary $\mathcal{M} = \{\}$, global counter $t = 1$, step limit L, a flag $delayed$ to indicate whether feedback is delayed.

repeat
  Initialize a counter (i.e., step) $l = 0$ and observe state s_l
  while s_l *is not terminal and* $l < L$ **do**
    Execution Loop
      Exploration Loop $k \in K$ **do**
        Compute activation values $A_i(t)$ of instances $(k_i, T(i))$ by Eq: (1)
        Compute retrieval probabilities $P_i(t)$ by Eq: (2)
        Compute blended values $V_k(t)$ corresponding to k by Eq: (3)
      end
      Choose an action a corresponding to option $k_l \in \arg\max_{k \in K} V_k(t)$
    end
    Take action a, move to state s_{l+1}, observe s_{l+1}, and receive outcome u_{l+1}
    Store t into instance corresponding to selecting k_l and achieving outcome u_{l+1} in $\mathcal{M}$
    If $delayed$ is true, update outcomes using a *credit assignment* mechanism
    $l \leftarrow l + 1$ and $t \leftarrow t + 1$
  end
until *task stopping condition*

Algorithm 1: Instance-Based Learning Process

3. Human-Autonomy Team Defense

3.1. Paradigm as HAT

A definition of HAT provided by the systematic review of O'Neill, McNeese, Barron, and Schelble (2022) states that a team includes autonomous agents as individual members who are recognized and seen as performing a unique role on the team with humans. Their work identified both interdependence ("acting with other member activities and outcomes") and agency ("involving independence of actions among autonomous agent members") as two major criteria for autonomous agents to be seen as teammates rather than tools. The first criterion speaks to the nature of the interaction, and the second speaks to the perceived capacity of the agent.

Interdependence. Interdependence can arise from the interrelatedness of the task for each team member, the team structure that requires interaction among team members, or the common goal and shared outcomes (Johnson & Johnson, 1989; Ramamoorthy & Flood, 2004). In TDG, human participants are monetarily rewarded based on the outcomes achieved by the human participant and the autonomous agent together. The collaboration structure also creates an interdependence that necessitates an exchange of information (here, the intention of actions) between the autonomous agent and the human, similar to what happens in human-human cyber-teams.

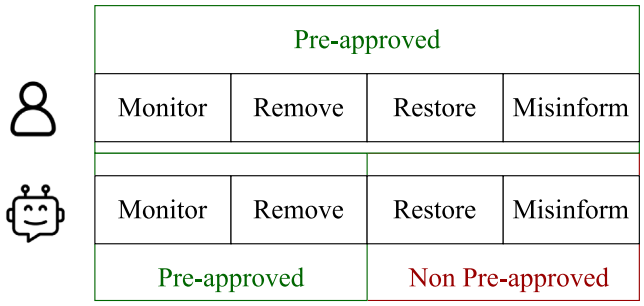


Fig. 1. Differences in autonomy of action between human and agent teammates. The autonomous teammate has to ask the human for approval of the *Restore* and *Misinform* actions.

Agency. One method of measuring the 'agency' of an autonomous agent is the Parasuraman Levels of Automation (LOA) paradigm (Parasuraman, Sheridan, & Wickens, 2000). O'Neill et al. (2022) assert that an autonomous agent in a HAT should at least have partial autonomy, i.e., they can perform actions for themselves instead of suggesting alternatives for humans. In the TDG, the level of autonomous agents falls between level 5, where the agent proposes a course of action but will not enact its decision without human approval ("non-pre-approved" actions), and level 7, where the agent chooses and enacts its course of action, while notifying the human teammate.

3.2. Team Defense Game (TDG)

We designed the Team Defense Game (TDG), an online cyber defense game developed to study how humans make decisions in collaboration with an automated agent to defend a network from cyberattacks. TDG is an extension of the interactive defense game (see Prebot, Du, & Gonzalez, 2023) adapted from the CAGE challenge (Standen, Lucas, Bowman, Richer, Kim et al., 2021), a cyber defense competition created to foster autonomous cyber defense research. In TDG, human participants play the role of cyber analysts. They are tasked with protecting the computer network of a fictitious manufacturing company against external malicious activity. Finding critical security incidents among a large number of false alerts generated from separate security products is cognitively demanding and stressful, often leading to frustration and performance degradation (Ban, Samuel, Takahashi, & Inoue, 2021; Dykstra & Paul, 2018; Nobles, 2022). To combat fatigue, human participants are paired with an artificial teammate, an autonomous cyber defense agent, who can make decisions and partially act independently to collaborate with the human defender in a team. Human and autonomous agents must collaborate efficiently to monitor the network, detect suspicious activity, and take appropriate actions to protect the network. The TDG provides human participants with a user interface to observe and analyze network events while interacting with an autonomous teammate protecting the same network and to supervise the actions of the teammate.

In human-human cyber teams (e.g., Cyber Protection Team, Incident Response Team, Security Operation Centers), analysts are provided with a set of pre-approved actions that they can execute without consulting their superior (Boyarchuk, Khudyntsev, Lebid, & Trofymchuk, 2021). If they consider that the situation requires actions that they cannot perform without approval from a superior, they need to submit their intention to their chain of command for validation before acting. It is a common practice to limit or expand the level of autonomy of human analysts through technical controls in the form of group memberships, password protection, firewalls, and even physical access. The idea that an autonomous teammate would have its level of autonomy controlled in a similar way makes it very similar to the technical controls that teams already use for human team members (Hauptman, Schelble,

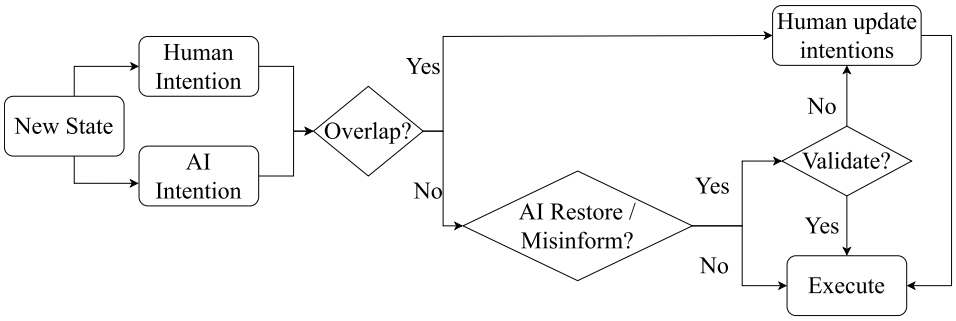


Fig. 2. Interaction flow.

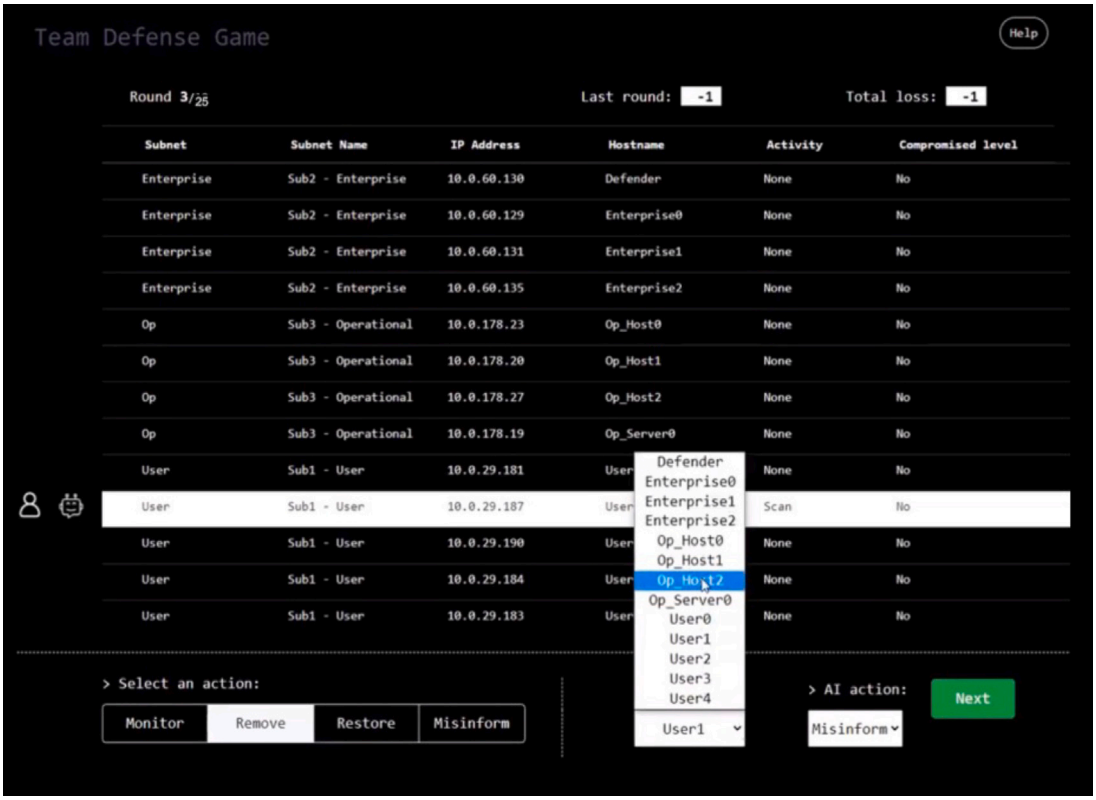


Fig. 3. User Interface example of the TDG.

McNeese, & Madathil, 2023). In TDG, we consider the autonomous agent a cyber analyst with a restricted set of pre-approved actions. As with its human counterparts in real-world settings, the autonomous agent can select an action that is not pre-approved but must submit for approval or modification to its (human) teammate. As shown in Fig. 1, the HAT consists of one human and one autonomous agent, each having a set of pre-approved actions. The agent takes the role of a low level analyst, and the human is the superior in the hierarchy. The agent's pre-approved actions are a subset of the human's pre-approved actions.

As shown in Fig. 2, in each step of the game, the decision to be made by a human analyst is a target (i.e., what computer or server to protect immediately) and an action to take on that target, and the same applies to the autonomous agent. Both make their decision without knowing the intention of their teammate. The human is presented with the agent's intention after he submits his intention. If the agent's intention involves one of the "pre-approved" actions (i.e., Monitor or Remove — see Fig. 1), it is "approved" by default, and the action is performed without the human's involvement. If the agent chooses to perform a "non-pre-approved" action (i.e., *Restore* or a *Misinform*), the human is prompted to validate or modify this intention by changing either the selected action or the target. If the human and autonomous agent select the same target, the human is prompted to resolve this overlap. To do so, they must modify either one of the intentions (human's or agent's). The human decides when to proceed to the next step of the game which will execute the two actions selected by the human and the agent. The agent's decision is executed first. The order in which actions are executed does not affect the effects of these actions.

As an example shown in Fig. 3, the human decides to take the Remove action on host User1; the agent decides to take the Misinform action on the same host User1. In this situation, the human participant is prompted to resolve the conflict. The human participant is allowed to change the agent's action, target, or his own action or target. The team will not be able to move to the next step until the overlap of the human's target and the agent's target is resolved.

3.3. Cyber scenario

As a simulated testbed, TDG allows customization of cyber scenarios. Each scenario consists of a simulated network, an adversary, and a defense team. TDG supports networks with arbitrary topologies

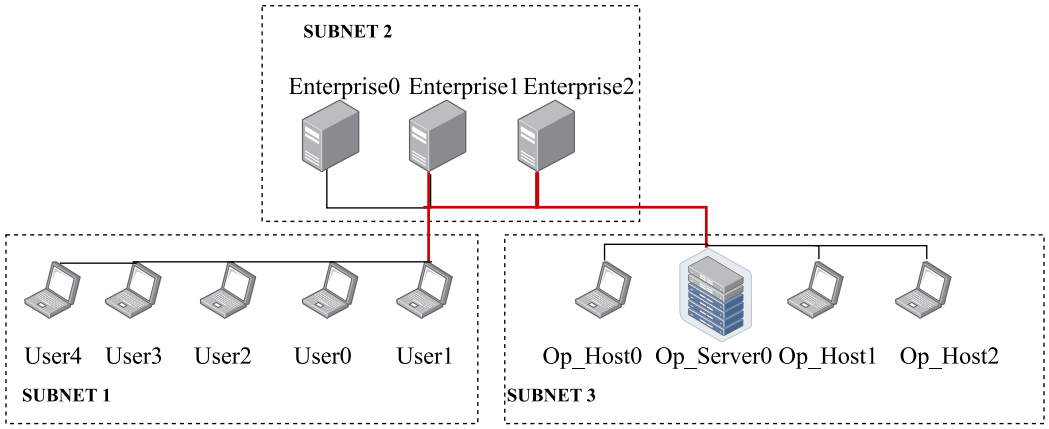


Fig. 4. Network configuration consisting of three subnets.

Table 1
Defender's loss caused by attacker's access and the cost of defense actions.

Event or Action	Subnet	Cost (Points)
Attacker has administrator access on a Host	Subnet 1	−0.1
Attacker has administrator access on a Host	Subnet 2	−1
Attacker runs IMPACT attack on Operational Server	Subnet 3	−10
Defender Restore an Host	–	−1
Defender Misinform (deploy a decoy)	–	−0.5/step (max 5 steps)
Defender disturb a regular User	–	−0.5

and adversaries with various attack capabilities and strategies. In this section, we introduce the cyber scenario used in the human-subject experiment.

In this scenario, the network is a simplified version of a common corporate network. As shown in Fig. 4, the simulated computer network consists of 13 hosts (9 computer hosts and 4 servers) distributed over three subnets. Subnet 1 consists of user hosts that are not critical. Subnet 2 consists of enterprise servers designed to support user activities on Subnet 1. Subnet 3 contains the critical operational server (Server 0), which maintains a service that is key to the operations of the system owners and some other operational hosts. The goal of the adversary is to navigate through the network to the Operational Server (Op_Server0) to steal valuable information and disrupt the network, which incurs a large cost to the defenders (see Table 1). The adversary algorithm follows a deterministic strategy that assumes prior knowledge of the network layout and is efficient, as it takes the shortest route to the operational server (see the red path in Fig. 4). Following this attack path, the adversary enters the network from a staff computer (i.e., User1) on the first subnet and makes its way to the critical Op_Server0 by gaining administrator access to every host on their way. Each host on which an attacker has administrator-level access is costly for the defenders (represented in a loss of points).

The goal of the defense team is to minimize the number of points lost. To understand the costs associated with events and actions, the TDG interface provides human participants with a table representation of the computer network. Each element of the network is represented as a row in the table, associated with some static parameters (name, IP address, subnet to which it belongs) and dynamic parameters that represent the state of infection of the host (*Compromise level*) and the last *Activity* detected by the system, such as scans or exploitation attempts performed by the attacker on this host. To add realism and complexity to the task, regular network activity is generated by simulated regular users who perform random scans. Specifically, a regular user scanned a randomly-chosen asset in approximately 5 steps out of the 25 steps in each episode. Therefore, defenders must understand the observable activity and compromise levels to discriminate suspicious activity and anticipate future actions of attackers. Based on these observable elements, Human participants and Agents in the defense team are given 4 possible actions to choose from in each step: (1) *Monitor* the network (i.e., do nothing), (2) *Remove* user-level adversary access to hosts, (3) *Restore* a system back to a standard configuration which will remove exploited privilege levels, (4) *Misinform* to deploy a decoy which can engage with the attacker to disrupt its operations and delay its progress. *Restoring* a system is guaranteed to remove adversary activity, but it is assumed that restoring a system disrupts the activities of legal users on that system. *Misinforming* with the honey service also requires careful quarantine and consumes computing resources. Thus, using these two commands is costly for the defense team.

The scenario consists of 7 *episodes*, each composed of 25 *steps*. To evaluate learning across episodes, at the end of each episode, the network and the attacker's progress are reset to the same initial state. At each step, the defense team takes two defense actions (one by the human and one by the autonomous agent), followed by an action by a regular user of the network. To add realism and complexity to the task, regular network activity is generated by simulated regular users who perform random scans. Specifically, a regular user scanned a randomly chosen asset in approximately 5 steps out of the 25 steps in each episode. Therefore, defenders must understand the observable activity and the levels of compromise to discriminate suspicious activity and anticipate future actions of attackers. Finally, the adversary algorithm observes the network and enacts an attack action. The cost to the defenders at each step is calculated based on the actions taken by the HAT as shown in Table 1, including the cost of losing hosts of various significance and the expense of *Restore* and *Misinform* actions. The cost of the previous round and the cumulative loss are displayed to the human participants. The goal of the defense team is to minimize cumulative costs.

4. Autonomous Defender Agents

In this section, we introduce the three types of autonomous agents used in the TDG as partners with a human player, including: a Cognitive agent, a Heuristic agent, and a Random agent.

4.1. Cognitive agent

The Cognitive agent determines the actions to take according to the IBLT algorithm introduced in Section 2.2.1. This agent makes decisions under the guidance of cognitive principles and has been shown to make defense decisions similar to humans (Gonzalez, 2024). The instances represent each decision made and are structured with the following three elements:

State: instance states are constructed to resemble the information that a human defender would have access to and use to determine the action they take. Specifically, there are two attributes for each host or server, representing the observed activity and the known compromised status of that host. The order of *(Activity, Compromised Status)* pairs for each host is fixed to encode the identity of each host, i.e., the *Host name*, *IP address*, and *Subnet*. The *Step Index* slot is included to indicate the step counter within each episode.

Action Space: The decision is for the Cognitive agent to choose a host to protect and the tool to protect it. Each action consists of a host and a command in the format of *cmd host*. The action space consists of each of the four actions that target each host in the network for a total of 40 possible actions.

Utility: The utility given to the Cognitive agent is the loss of the team from the last step in each step. As shown in Table 1, team loss is the sum of loss caused by attacker's access to the network and the cost of defense actions.

The Cognitive agent accumulates memory of defense decisions and their utility while defending a network against the adversary. The amount of noise added during instance activation computation is set to the default value 0.25. The rate at which activation for previously experienced instances in memory decay with the passage of time is set to the default value 0.5. The memory retention limit is set to 250000 instances. Note that the Cognitive agent is essentially a zero-parameter model, as the decay and noise parameters were not manipulated, fitted, or adapted in any way to its partner's behavior. During training, the cognitive agent gathers experience in an individual version of the TDG, where its actions are automatically approved. This training period lasts 500 episodes, after which the model achieves an average cumulative episode loss of 68.95 ± 7.23 (see Table 1). When the agent makes a decision, during training and the main experiment, it does so by predicting the expected utility of each action that is available to it according to (3), and selecting the maximum utility action. During the training period, the agent successfully learns to defend against the attacker by selecting actions that maximize utility. Additionally, the memory capability of the IBL model allows it to continuously learn and change its action selection based on the utility that the team observes. It is important to note that the Cognitive agent does not directly observe or model its human teammate's specific actions. Rather, it observes: (1) the joint utility achieved by the team, which integrates both players' contributions, and (2) the resulting network states after both players act. Through its instance-based memory, it builds a repository of situation-action-outcome experiences, with more recent and frequent experiences having higher activation according to Eq. (1). This is how the agent indirectly incorporates human influence into its decision-making process without explicit teammate modeling.

Input: Current network state $NetworkState$
Output: Selected defense action
$options \leftarrow \emptyset$
$exploitedHosts \leftarrow$ `GetHostsWithStatus(`$NetworkState$, *"Exploited"*`)`
$privEscedHosts \leftarrow$ `GetHostsWithStatus(`$NetworkState$, *"PrivEsced"*`)`
$attackPathHosts \leftarrow$ `GetHostsOnAttackPath(`$NetworkState$`)`
$options$.Add("Monitor")
if $exploitedHosts \neq \emptyset$ **then**
  $mostImportantExploited \leftarrow$ `GetMostImportantHost(`$exploitedHosts$`)`
  $options$.Add("Remove " + $mostImportantExploited$)
end
if $privEscedHosts \neq \emptyset$ **then**
  $mostImportantPrivEsced \leftarrow$ `GetMostImportantHost(`$privEscedHosts$`)`
  $options$.Add("Restore " + $mostImportantPrivEsced$)
end
if $attackPathHosts \neq \emptyset$ **then**
  $relevantHost \leftarrow$ `GetMostImportantHost(`$attackPathHosts$`)`
  $options$.Add("Misinform " + $relevantHost$)
end
$selectedAction \leftarrow$ RandomChoice($options$)
return $selectedAction$
Function `GetMostImportantHost(`$hosts$`):`
  Sort $hosts$ by subnet importance (Subnet 3 > Subnet 2 > Subnet 1)
  return first host in sorted list

Algorithm 2: Heuristic Agent Decision Making

4.2. Heuristic agent

As a baseline to compare the Cognitive agent, we designed the Heuristic agent that was formalized as a set of rules, which are applied according to the state of the network. The algorithm of the Heuristic agents is shown in Algorithm 2.

At each step, the Heuristic algorithm identifies compromised hosts and randomly selects contextually appropriate defense actions based on the current state of the network. These heuristics assume that an agent has full knowledge of the network structure, the losses associated with each action taken by adversaries, and knows which actions will have the best chance of preventing the progress of the attacker. Furthermore, the performance of Heuristic agents share connections to the expected behavior of reinforcement learning (RL) after long periods of training (Sutton, Barto et al., 1998). After finding the optimal or near-optimal decision strategy during training, the RL models show stable behavior.

Both agents, Cognitive and Heuristic, are dynamic, but the Heuristic agent relies on pre-defined rules, which do not change over the course of an episode of the task. Also, the Heuristic agent is as competent as the cognitive agent when performing the task in isolation, achieving an average cumulative episode loss of (66.20 ± 6.43). However, the Heuristic and Cognitive agents differ fundamentally in their underlying algorithms. The Heuristic agent uses fixed rule-based decision-making, selecting randomly from predetermined correct options based on the network state, while the Cognitive agent employs the IBLT algorithm to accumulate experiences and adapt its decision-making over time according to their cognitive mechanisms.

To illustrate the behavior of the Cognitive and Heuristic agents, Appendix (Behavior Analysis of the Autonomous Agents) shows the distribution of target selection patterns across simulation steps. The appendix suggests that the dynamics of the Cognitive and Heuristic agents are similar but also differ in concrete ways in which the agents

address the attacker's progression over the User, Enterprise, and Operational subnets. This suggests that the Heuristic agent is a good baseline comparison to the Cognitive agent.

4.3. Random Agent

The key similarities between the Heuristic and Cognitive agents are their overall competence in the task and their ability to make decisions that directly depend on the network state of the cyber defense task. To get a more basic baseline, a Random agent was developed to evaluate teaming performance in the TDG, and to compare participants' perception of trustworthiness and cooperativeness.

The Random agent selects actions by choosing a resource as a target at random and then choosing an action to perform at random. All resources and actions have an equal probability of being selected by the random agent. As a result, the performance of the Random agent acting in isolation is considerably lower than that of the IBL or Heuristic models, achieving an average cumulative episodic loss of 121.39 ± 49.44. Due to this poor individual performance, the Random agent is not expected to achieve a high performance when paired with human participants, though results from this can clarify the relative improvements that are expected by using the Heuristic and cognitive agents.

Just as the Heuristic agent reflects the expected behavior of methods like RL, so does the Random agent. The Heuristic agent behaves similarly to an RL agent at the end of its training lifecycle, while the Random agent resembles an RL agent in the early stages of training. Early-stage RL agents typically exhibit roughly random action selection, which is often used in RL research to compare the performance of trained versus untrained agents (Sutton et al., 1998).

5. Experiment method

5.1. Participants

Participants were recruited through Amazon Mechanical Turk to participate in a cybersecurity study. The study was advertised to last about 60 min. The experiment took $M = 56.20 \pm 15.14$ Minutes on average. Participants received a base compensation of $6, and up to $12 in bonus payment based on their final score[1] for a total possible payment of $18. The average bonus payment was 11.41 ± 2.23. 156 participants (63 female, 90 male, 3 N/A) aged 22–65 years ($M = 39.49 \pm 9.35$) completed the study. 66 (27 female, 36 male, 3 N/A) were paired with the Random agent, 48 (16 female, 32 male) with the Heuristic agent, and 42 (20 female, 22 male) with the Cognitive agent. The different number of participants for each condition is caused by random assignment. We remove data from participants who did not fully complete the experiment and the post-experiment questionnaire. Although no formal attention checks were implemented, strict data cleaning was carried out. Participants with more than one missing value and participants whose completion time deviated by more than ±3 standard deviations from the mean were excluded from the analysis. We also excluded participants who showed signs of inadequate engagement, such as taking the same action throughout the game.

[1] As the score used in this experiment is negative (loss), the bonus payment was calculated by using the difference to the maximum possible loss and attributing 0.005$ per point: bonus=(*total loss*+1120)*0.005. 1120 is the maximum total loss for seven episodes

5.2. Procedure

First, participants had to complete a demographic questionnaire and provide informed consent. Then, they received instructions for the task and were asked to complete a short quiz to verify their basic understanding of the instructions. The participants had to correctly answer all the questions before moving on to the next step of the experiment. They received feedback on the accuracy of their responses and were allowed to modify their responses if they were incorrect. There was no limit to the number of attempts by participants to answer the questions correctly. However, for control purposes, we recorded the score of their first attempt and the number of times they tried to answer the questions. The participants then watched a video introduction to the TDG explaining the interface, the game controls, and the dynamics of an episode. The participants were then led to a practice session consisting of 1 short episode (that is, a game) of 10 steps. The practice episode was intended to familiarize participants with the interface and game controls and present them with situations in which they must deal with supervision, overlap, and misinformation.

Following the practice session, each participant was randomly assigned to work in conjunction with one of the three types of autonomous agents. The participants performed the main task, which consisted of 7 episodes with 25 steps each. No time restrictions were imposed. The experimental conditions were constant throughout the episodes, which means that each participant played 7 episodes with the same autonomous teammate. The initial state of the network was the same for all participants and for each episode. Subsequently, the participants completed a post-experiment survey consisting of three parts: (1) a collaboration survey, (2) a trust survey, and (3) a background survey about their experience in computer science and cyber defense. Finally, the participants received their final score and were dismissed.[2]

5.3. Dependent measures

A summary of the metrics, their units and their description is shown in Table 2.

5.3.1. Team performance

We measured *team performance* with objective metrics: (1) *Loss*: the average cumulative episodic loss; (2) *Recovery time*: the average number of steps per episode that the adversary successfully impacts the operational server (that is, how long the team takes, on average, to stop an attack occurring on the operational server).

5.3.2. Collaborative process metrics

To evaluate how human and autonomous team members synchronize each other's activities in the interdependent team task, we measure their collaborative process in three situations: (a) *Overlap*, (b) *Supervision*, (c) *Backup*, which have the greatest potential to impact team performance.

Overlap. Overlap refers to situations in which the human player and the autonomous agent choose the same target. A high number of overlaps could suggest a lack of coordination. We measure (1) *Frequency of Overlap*: the number of overlap cases per episode (2) *Adjustment Rate*: the rate of two possible types of adjustment human participants could use to resolve the overlap: adjust their own action or adjust the agent's action. Self-adjustment demonstrates the willingness of human participants to adapt to the autonomous agent for cohesive collaboration.

[2] The experimental instructions, quizzes, and surveys, along with the data and analysis scripts, can be accessed on Open Science Framework: https://osf.io/xk624/.

Table 2
Metrics summary.

Metric	Unit/Scale	Description
Team Performance Metrics		
Loss	Points	Average cumulative episodic loss (negative values indicate points lost). Lower values (closer to zero) indicate better performance.
Recovery Time	Steps	Average number of steps per episode required to stop an attacker's impact on the operational server. Lower values indicate faster recovery and better performance.
Human–Agent Interaction Metrics		
Frequency of Overlap	Proportion	Rate of instances where human and agent selected the same target per episode (0–1 scale). Lower values indicate better coordination.
Adjustment Rate	Proportion	Rate at which humans adjusted their own actions (vs. the agent's) when resolving target conflicts (0–1 scale). Higher values indicate human adaptability.
Frequency of Supervision	Proportion	Rate of instances per episode where the agent required human validation for non-pre-approved actions (0–1 scale).
Agreement Rate	Proportion	Rate at which humans approved the agent's actions without modification during supervision (0–1 scale). Higher values indicate greater trust.
Frequency of Multiple Breaches	Proportion	Rate of instances per episode when multiple hosts were compromised simultaneously (0–1 scale). Lower values indicate better defense.
Backup Rate	Proportion	Rate at which humans attempted to recover breached hosts when multiple breaches occurred (0–1 scale). Higher values indicate better backup behavior.
Human Effort and Efficiency Metrics		
Human Effort	Actions/Episode	Average number of active actions (Remove, Restore, Misinform) taken by humans per episode. Lower values may indicate less human workload.
Human Efficiency	Points/Action	Loss reduction divided by human effort. Higher values indicate humans achieved better results with fewer actions.

Supervision. Supervision refers to the cases where the agent intends to take an action that requires the validation of its human teammate (i.e., *Restore* and *Misinform*). We measure (1) *Frequency of supervision*: the number of times per episode the Agent needs a validation of their action from the human; (2) *Agreement Rate*: the rate of times the human participant allows the agent to execute its intended action without modification, out of the total number of times the Agent needs a validation of their action from the human. A high *Agreement Rate* could indicate that the agent's decision making aligns well with the human participant's judgment and is trusted.

Backup. The challenging task of defending multiple hosts legitimately calls for backing-up behavior. We measure (1) *Frequency of Multiple Breaches*: the rate of times that more than one host in the network are compromised; (2) *Backup Rate*: the rate of times that the human participant attempts to recover a breached host, out of the total number of times there are multiple breaches in the network. A high *Frequency of Multiple Breaches* shows a high demand for both members of the HATs to contribute and recover the compromised host. A high *Backup Rate* suggests that human participants can rise to the challenge and support their agent teammate when necessary.

5.3.3. Human Effort and Efficiency

Engagement (Sidner, Lee, Kidd, Lesh, & Rich, 2005) is another key process that underlies how effectively autonomous agents can interact with human partners (Holroyd, Rich, Sidner, & Ponsler, 2011; Sidner, Lee, & Lesh, 2003). To evaluate whether human participants are actively and effectively engaged in teamwork, we measure (1) *Human Effort*: The frequency of humans taking active actions (Remove, Restore, Misinform) rather than passively monitoring per episode; and (2) *Human Efficiency*: Loss reduction divided by the total human effort during the episode. The loss reduction is calculated by the difference between the observed loss and the maximum loss over all participants in the experiment, to result in a positive value for our Efficiency metric.

5.3.4. Human perception of the autonomous agent

In the post-experiment questionnaire shown in Table 3, we measured *Cooperativeness* and *Trustworthiness*. The cooperativeness and trustworthiness survey questions were inspired by previous studies on automation trust, including discussions in Glikson and Woolley (2020), Schelble et al. (2022). We measured the perceived cooperativeness of the autonomous teammate through a home-made survey composed of 6 items, each rated on a 5-step Likert scale. For the trustworthiness survey we kept all 6 items of Merritt's trust scale (Merritt, Heimbaugh, LaChapell, & Lee, 2013) and adapted the questions to make reference to the "teammate" rather than the "automation". This survey is also based on a 5-step Likert scale. Given the novelty of the current study, where automation is used as a teammate, we could not rely on well established metrics. We used these questions simply to have a subjective metric of cooperativeness and trust, in addition to the objective metrics.

6. Results

In this section, we present the experimental results comparing the three agent types (Random, Heuristic, and Cognitive) in human-autonomy teaming for cyber defense. We analyze team performance metrics, collaborative process patterns, human effort indicators, and participant perceptions of agent trustworthiness and cooperativeness. Table 4 summarizes the descriptive statistics in each of the three agent conditions, which will discuss and test for their significance.

6.1. Team performance

Fig. 5 presents the average loss of the team and the average recovery time of the team per episode for all HATs under the three conditions. The minimum possible loss per episode is 0 and the maximum is 160. As shown in Table 4, the HAT loss is largest when humans are paired

Table 3
Cooperativeness and Trustworthiness Questionnaire. Participants had to rate each item on a 5-step Likert scale.

Cooperation	My teammate and I coordinated our actions well together My teammate and I coordinated our actions better over the course of the episodes. My teammate and I contributed equally to the team performance I had to carry the weight to make the team better My teammate perceived accurately what task I was trying to accomplish. I was able to understand and predict what task my teammate was trying to accomplish.
Trust	I believe my teammate is a competent performer. I trust my teammate. I have confidence in the choices taken by my teammate. I can depend on my teammate. I can rely on my teammate to behave in consistent ways. I can rely on my teammate to do their best whenever I validate its decision.

Table 4
Descriptive Statistics: Mean and Standard Deviation of HATs with three types of agents.

Metric	Random	Heuristic	Cognitive
Team Performance			
Loss	M = −79.69 SD = 49.11	M = −59.69 SD = 28.54	M = −52.85 SD = 27.42
Recovery time	M = 3.92 SD = 3.89	M = 2.15 SD = 2.09	M = 1.45 SD = 1.89
Human–Agent Interaction			
Frequency ofOverlap	M = 0.11 SD = 0.08	M = 0.08 SD = 0.08	M = 0.078 SD = 0.09
Adjustment rate(of human actions)	M = 0.14 SD = 0.24	M = 0.29 SD = 0.34	M = 0.21 SD = 0.33
Frequency ofSupervision	M = 0.49 SD = 0.10	M = 0.33 SD = 0.11	M = 0.37 SD = 0.13
Agreement Rate	M = 0.35 SD = 0.012	M = 0.74 SD = 0.017	M = 0.73 SD = 0.02
Frequency ofMultiple Breaches	M = 0.41 SD = 0.02	M = 0.14 SD = 0.021	M = 0.14 SD = 0.013
Backup rate	M = 0.20 SD = 0.02	M = 0.17 SD = 0.02	M = 0.11 SD = 0.03
Human Effort and Efficiency			
Human Effort	M = 10.67 SD = 7.13	M = 9.28 SD = 7.18	M = 8.38 SD = 7.52
Human Efficiency	M = 16.80 SD = 27.02	M = 29.82 SD = 38.32	M = 39.40 SD = 46.53

with Random agents, comparatively lower with Heuristic agents, and minimal when humans are paired with Cognitive agents. The HAT recovery time follows a similar pattern, with a longer time to recover a breached host when humans are paired with Random agents, a shorter duration with Heuristic agents, and the shortest duration when paired with Cognitive agents. These observations strongly suggest that Cognitive partners in HATs are human's most effective team collaborators. It is worth noting that Fig. 5 shows a sudden drop in loss from episode 4 to episode 5. Upon inspecting the human data, we found that two participants were able to stop the adversary earlier in episode 5 compared to episode 4. This early intervention limited the adversary's opportunities to launch Impact actions, reducing the average loss by approximately 10 points—roughly equivalent to preventing one Impact action. This finding aligns with the observed decrease in recovery time from episode 4 to episode 5.

A two-way mixed measures ANOVA was performed to compare the team loss of HATs with three types of agents over 7 episodes. The results indicated a significant main effect for the type of agent, $F(2, 152) = 11.037, p < .05, \eta_G^2 = 0.086$. There was no significant interaction between the type of agent and the episode, $F(11.26, 855.40) = 0.938, p = 0.504, \eta_G^2 = 0.005$. There was also no significant effect of the episode $F(5.63, 855.40) = 0.938, p = 0.821, \eta_G^2 = 0.001$. Post hoc tests using Tukey's HSD indicated that HATs with Cognitive agent achieve a significantly lower loss than HATs with Heuristic agent ($p = 0.029$) and lower than Random agent ($p < .001$).

Similarly, the two-way mixed measures ANOVA for the recovery time of HAT indicated a significant main effect for the agent type, $F(2, 152) = 18.297, p < .05, \eta_G^2 = 0.117$ but no effect of the interaction between agent type and episode, $F(11.38, 864.78) = 0.95, p = 0.744, \eta_G^2 = 0.003$; and no effect of episode, $F(5.69, 864.78) = 0.95, p = 0.313, \eta_G^2 = 0.005$. Post hoc testing using Tukey's HSD indicated that HATs with a Cognitive agent achieve a significantly shorter recovery time than HATs with a Heuristic agent ($p = 0.007$) and shorter with a Random agent ($p < .001$).

6.2. Conflict resolution

As illustrated in Table 4, overlap occurs more often in the *Random* condition. A two-way mixed measures ANOVA was conducted to compare the overlap frequency of HATs with three types of agents in 7 episodes. The results confirm a significant effect of the agent type, $F(2, 152) = 4.912, p = 0.009, \eta_G^2 = 0.046$, and Post hoc tests using Tukey's HSD indicated that HATs with Random agent indeed observe significantly more overlap situations than HATs with Cognitive agent ($p < .001$) and HATs with Heuristic agents ($p < .001$).

Humans appear to mostly modify the action of agents when there are overlaps, even when they are paired with *Heuristic* and *Cognitive* agents. A two-way mixed measures ANOVA was performed to compare the rates of Human self-adjustment with three types of agents over 7 episodes. The results indicated a significant effect for the type of agent, $F(2, 65) = 4.607, p = 0.013, \eta_G^2 = 0.084$, and a significant effect for the episode, $F(6, 390) = 3.670, p = 0.001, \eta_G^2 = 0.014$. There was no significant interaction between the type of agent and the episode, $F(12, 390) = 0.662, p = 0.788, \eta_G^2 = 0.005$. As shown in Table 4, post hoc testing using Tukey's HSD indicated that in overlap situations, when paired with a Heuristic agent, humans adjust their own decision significantly more than when paired with Cognitive ($p = 0.015$) and Random ($p < .01$)). When paired with Cognitive agent, humans also tend to adjust their own decision more than when paired with Random ($p = 0.14$). Further analysis revealed that about 80% of the time, they tend to resolve the conflicts by changing the agent's command to *Monitor*.

6.3. Supervision

As shown in Table 4, random agents require the most supervision. A two-way mixed ANOVA was performed to compare the frequency of HAT supervisory cases with three types of agents in 7 episodes. The results indicated a significant main effect for the type of agent, $F(2, 152) = 76.811, p < .001, \eta_G^2 = 0.277$. There was no significant interaction between the type of agent and the episode, $F(11.49, 872.98) = 0.957, p = 0.133, \eta_G^2 = 0.011$. There was also no significant effect of the episode$F(5.74, 8872.98) = 0.957, p = 0.424, \eta_G^2 = 0.004$. Post hoc testing using Tukey's HSD indicated that HATs with Random agents

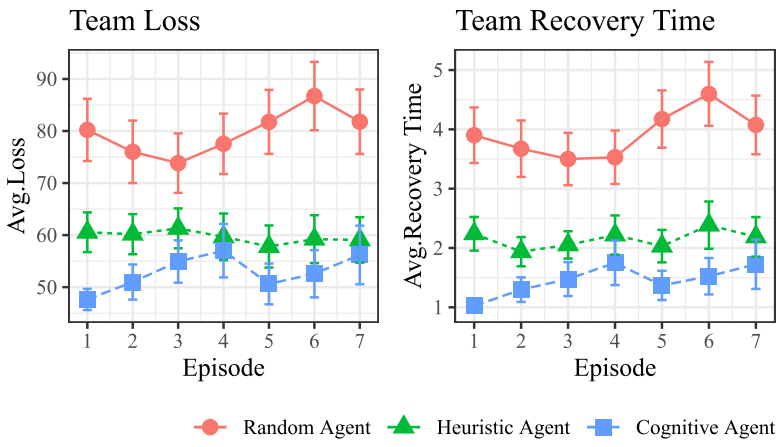


Fig. 5. Evolution of team performance across the 7 episodes of the experiment composed of Team Loss per episode (left) and Team Recovery Time (right).

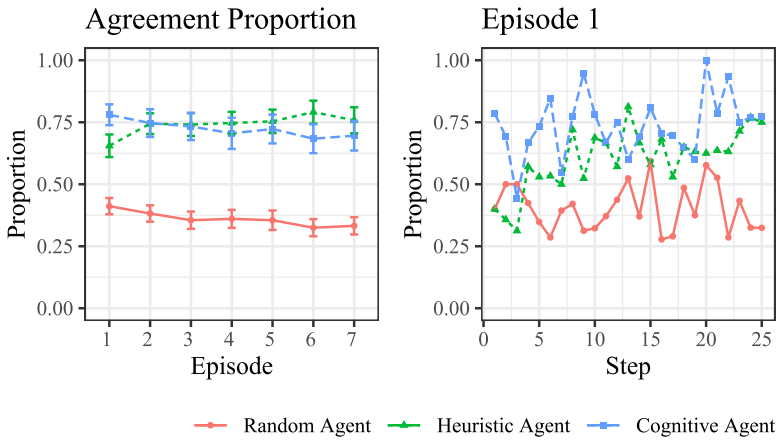


Fig. 6. Evolution of the rate of Human's agreement with Agent's intention during supervisory situations, throughout the first episode (right) and the entire experiment (left).

require significantly more supervision than HATs with Heuristic agents ($p < .001$) and Cognitive agents ($p < .001$).

The left panel of Fig. 6 presents the agreement rate throughout 7 episodes. Humans agreed significantly more with the Cognitive and Heuristic agents starting from the first episode. The agreement rate increase in the Heuristic condition is probably due to the consistency and, thus, predictability of the Heuristic agents. A two-way mixed measures ANOVA was conducted to compare the rate of human agreement of HATs with three types of agents across 7 episodes. There was a significant interaction between agent type and episode, $F(8.73, 663.49) = 0.728, p < .05, \eta^2_G = 0.254$. Post hoc testing using Tukey's HSD indicated that Cognitive agents and Heuristic agents get significantly more agreements than random agents. Further inspection of the agreement rate within episode 1 is presented in the right panel of Fig. 6. By the end of the first episode, the agreement rate increased to approximately 75% in the Cognitive and Heuristic condition and dropped to approximately 35% in the random condition, which indicates that humans are able to observe the competencies of teammates rapidly (Abele, Ellemers, Fiske, Koch, & Yzerbyt, 2021) during the initial exploration episode.

Closer examination of the agreement rate regarding the Restore vs. Misinform actions showed that human participants are able to differentially place their trust in the autonomous agent intentions that vary in reliability. We observed that, independently of the type of teammate, humans tend to agree more with the Restore action (respectively 81%, 80%, and 49% in Heuristic, Cognitive, and Random HATs), which has a deterministic benefit to the team, compared to the Misinform action (71%, 70% and 21% of agreement) that deploy traps which might not catch the adversary successfully every time. At the same time, human participants have the tendency to merge their trust across actions despite their differences in the actions' reliability. Overtime, the consistent and accurate Restore actions from the Heuristic agent earned itself more trust and led to a higher false agreement rate to its Misinform actions in later episodes.

6.4. Backup behavior

We consider a Backup to be a state of the environment in which there are at least two hosts that are compromised simultaneously. As shown in Table 4, situations requiring backup behavior are more common in HATs with Random agents than with Cognitive or Heuristic agents. A 1-way ANOVA was performed to compare the frequency of multiple breaches between HATs with the three types of agents. The results confirmed a significant effect of the agent type, $F(2, 147) = 46.85, p < .001, \eta^2_G = 0.174$. A post-hoc test using Tukey's HSD indicated that HATs with Random agents get significantly more multiple breach situations than HATs with Cognitive ($p < .001$) and Heuristic ($p < .001$).

Also shown in Table 4, Random agents encounter more backup states from their human teammates, with a higher rate of times (20%) when there is more than one breached host. Humans are not good at identifying the need to backup the agents, especially when the agents (i.e., Heuristic and Cognitive agents) appear capable. A two-way mixed measures ANOVA was conducted to compare the rate of backup behaviors of Humans in HATs with three types of agents across 7 episodes. The results indicated a significant effect for the agent type, $F(2, 680) = 18.621, p < .001, \eta^2_G = 0.093$. There was also a significant effect of the episode, $F(1, 680) = 5.018, p = 0.025, \eta^2_G = 0.065$. The backup rate decreases overtime. There was no significant interaction between the type of agent and the episode, $F(2, 680) = 1.758, p =$

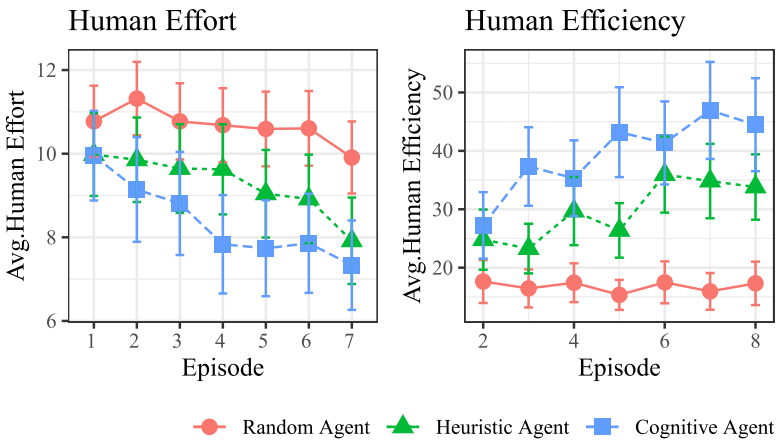


Fig. 7. Evolution of Human Effort (left) and Human efficiency (right) across the experiment.

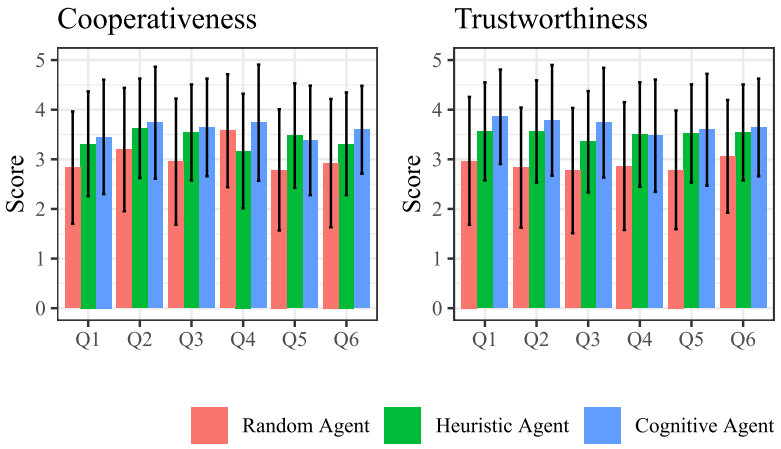


Fig. 8. Left: Cooperativeness Likert Scale, Right: Trustworthiness Likert Scale (1 - Strongly disagree, 2 - Disagree, 3 - Neither agree or disagree, 4 - Agree, 5 - Strongly Agree) - see Table 3.

$0.173, \eta_G^2 = 0.002$. Post hoc testing using Tukey's HSD indicated that Humans paired with a Cognitive agent provide significantly less backup to their teammates than Humans paired with a Heuristic agent ($p < .001$) and a Random agent ($p < .001$). Humans contribute the most when paired with Random agents(20%); however, not enough to compensate for the deficiency of the agents.

6.5. Effort and Efficiency

Fig. 7 left panel shows a decrease in the frequency of humans taking active actions (Remove, Restore, Misinform) rather than passively monitoring. A two-way mixed measures ANOVA was conducted to compare human effort in HATs with three types of agents across 7 episodes. In fact, there is a significant effect of episode $F(3.46, 526.09) = 5.463, p < .05, \eta_G^2 = 0.036$, where human effort decreases over episodes. There was no significant interaction between the type of agent and the episode, $F(6.92, 526.09) = 0.561, p = 0.79, \eta_G^2 = 0.003$ and there was no significant effect of the agent types $F(2, 152) = 1.801, p = 0.169, \eta_G^2 = 0.001$.

Fig. 7 shows a higher efficiency in participants paired with Cognitive agents compared to both Heuristic or Random agents. A two-way mixed measures ANOVA was conducted to compare the human efficiency of the participants in HATs with three types of agents across the 7 episodes. The results indicated a significant main effect for the type of agent, $F(2, 152) = 7.949, p < .001, \eta_G^2 = 0.145$, episode $F(3.46, 630.77) = 5.322, p < .001, \eta_G^2 = 0.037$ and interaction effect $F(8.30, 630.77) = 2.435, p = 0.012, \eta_G^2 = 0.092$. Tukey's HSD indicated that HATs with Cognitive agents achieve significantly higher human efficiency than HATs with Heuristic agents ($p = 0.0004$) and Random agents ($p < 0.001$).

Although participants' efforts did not show a significant difference between conditions, their efficiency did. This demonstrates the important relationship between autonomous agent strategy and the efficiency of humans in HATs. For HATs with Cognitive teammates, human participants achieved a high level of efficiency. This translates into a relatively low effort from the human participant compared to the improvement in performance observed. One possible explanation for this observed difference is that Cognitive agents are better suited to learning the individual play styles of participants, which can vary from more to less effortful, while the Heuristic and Random agents could not learn these differences. This is evidenced by the fact that in Cognitive and Heuristic HATs, participants begin with nearly the same efficiency on the first episode, which quickly diverges and remains higher for participants in Cognitive HATs.

6.6. Perception of agent cooperativeness and trustworthiness

As shown in Fig. 8, Cognitive and Heuristic agents are rated significantly more cooperative and trustworthy than the Random agent in general. A one-way ANOVA was conducted to evaluate human's perception of the cooperativeness and trustworthiness of the three

types of agents. The results indicated a significant effect of the type of agent on perceived cooperativeness, $F(2, 153) = 7.23$, $p<.05$, and trustworthiness $F(2, 153) = 10.125$, $p<.05$. 51.9% of participants in the Cognitive condition *Agree* or *Strongly Agree* that the Cognitive agent is cooperative and trustworthy, 49.8% in the Heuristic condition, and 32.9% in the Random condition.

The open-ended feedback in the questionnaire revealed that participants' perception of agents might be affected by their various expectations of the autonomous agent before the experiment. As shown in the following quote, a participant that partnered with a Heuristic agent expected some level of communication with the agent to facilitate decision-making and planning. The absence of such a feature caused frustration. This participant answered *Neither agree or disagree* to most of the survey questions.

> "It was challenging because you had to work with an AI that cannot communicate. Made it difficult to come up with a plan or strategize".

Another participant that teamed with a Random agent, on the other hand, has a high level of confidence in the agent's decision-making process. This participant selected *Agree* or *Strongly agree* to all of the survey questions.

> "I trusted the partner more than I trusted myself. I was glad it was AI, I assumed they were using an algorithm to make the decisions".

7. Discussion

Our results support three general implications for the design of autonomous agents that collaborate with humans in teams: (1) dynamic agents that emulate human-like cognitive processes are beneficial for HAT effectiveness, the benefit can be enhanced through providing explanations when the agents adjust their behavior; (2) competent agents are more trusted and can lead to better HAT performance but might cause over-reliance; (3) human trust in autonomous agents is dynamically shaped by their interactions during teamwork, and should be measured before, during, and after the collaboration.

Human-like cognition. Our first finding concerns the benefits of replicating human cognitive processes in autonomous cognitive teammates. From the comparison between HATs with Cognitive and Heuristic agents (the Cognitive agent achieves the same level of competence as the Heuristic agent after learning), we found that humans who partnered with cognitive agents became more efficient in reducing team loss over time. HATs with Cognitive agents also delivered better team performance. In the post-experiment questionnaire, the Cognitive agent scores slightly higher than the Heuristic agents on trustworthiness. However, there was no significant difference in human trust in autonomous agents. This suggests that human perception of teammate behavior may not be an effective metric for HAT performance. In supervision situations, human participants agreed to approximately 75% agent decisions from the first episode in both conditions. The agreement rates slightly increased over time in the Heuristic condition but slightly dropped in the Cognitive condition. In the presence of conflicts, humans tend to resolve the overlap by adjusting their own actions rather than the agent's actions. One explanation is that humans are less likely to adjust the behavior of the Heuristic agent more because of its consistent behavior. People can easily recognize the 'rules' it follows, making its actions predictable and understandable. In contrast, the Cognitive agent, despite its adaptability to the environment and the human teammate, exhibits less consistency and predictability. In summary, we found that autonomous agents endowed with human-like cognition can improve team effectiveness. However, agents must also employ certain behaviors, such as explanations, to maintain human trust.

Competence. Our second finding concerns the competence of autonomous teammates. From the comparison between the competent agents (i.e., Cognitive and Heuristic) versus the incompetent Random agent, we found that Humans demonstrate a significantly higher level of trust toward the competent agent, both behaviorally and subjectively. HATs with competent agents were significantly more effective; however, it is worth noticing that humans are not good at identifying the deficiencies and failures of competent agents. Most of the participants did not contribute in situations that require backup behavior. Consistent with previous empirical studies on system-wide trust (Walliser, de Visser, & Shaw, 2023), humans also present a tendency to apply trust broadly rather than specifically to each specific function of the agent. The consistent and accurate Restore actions from the Heuristic agent earned itself more trust and led to a higher false agreement rate to its Misinform actions in later episodes. In sum, the competence of autonomous agents is essential to gain human trust and achieve desirable HAT performance. However, for effective collaboration between humans and autonomous agents, it is crucial that agents actively signal when they require human assistance and transparently communicate the level of uncertainty (Demir et al., 2019) in their decision-making processes (Tomsett et al., 2020).

Trust measurement. Our last insight concerns the measurement of human trust in autonomous agents. First, we observed a significant effect of the episode on the rate of agreement, which echoes the findings of previous research that human trust can be affected by their interaction experience with agents (Kulms & Kopp, 2019). When establishing trust, affective and competency-based dimensions interact to dynamically shape human behavior toward the agent. In teamwork settings, the improved or deteriorated trust caused by previous episodes can affect the interaction in subsequent episodes and spiral into a success or vicious cycle that drags down both team confidence and performance over time. Therefore, it is crucial to regularly assess the different dimensions of trust to adapt the agent's behavior accordingly.

Second, we found that significant differences in agent cognitive ability and competence are more evident in behavioral measures than in self-reported scores. The post-experiment questionnaire showed that some participants had unrealistically high expectations of the agent, which later led to blind trust or great disappointment in their interaction with the agents. We also observed a discrepancy between the actual contribution and the perceived contribution. Many participants *Agree* or *Strongly Agree* that *"I had to carry the weight to make the team better"* while, actually, their agent made a greater contribution to the team. This could be due to the moral assumption that *"even if the autonomous agent gains more agency, humans remain responsible"* (Cummings, 2014).

Lastly, reinforcing the first two points of the discussion, our findings suggest that a high level of competency, when paired with strong predictability, can lead to over-reliance. We observed that humans tended to align more closely with the strategies of the heuristic agent, occasionally resulting in misplaced trust and reduced performance. In contrast, participants who worked with the cognitive agent were more critical of its decisions, which may indicate a more engaged and adaptive interaction. This suggests that cognitive agents could play a valuable role in dynamic trust calibration, helping to maintain team efficiency while preventing over-reliance. Measurement of behavioral adaptation, such as changes in human oversight, agreement with agent decisions, and response to unexpected agent actions, could then provide richer insights into trust dynamics than static self-reported trust ratings.

In summary, to better leverage human feedback on agents' trustworthiness and cooperativeness, we recommend refining subjective measures to distinctly capture different dimensions of trust, including propensity to trust (Hoff & Bashir, 2015; Schoorman, Mayer, & Davis, 2007), affective trust, and competency-based trust. Furthermore, trust levels should be unobtrusively monitored during teamwork using behavioral markers, while post-experiment questionnaires should clearly differentiate between taskwork and teamwork contributions.

7.1. Implications on HAT for cybersecurity

In the context of cybersecurity, the system-wide trust problem is more prominent. Autonomous cybersecurity systems tend to integrate more than one function, such as analyzing alerts, validating security controls, and resolving incidents based on standardized procedures and playbooks. Each task can be accomplished with different levels of trustworthiness and might evolve to a different level of risk. Thus, it is important to further investigate system-wide trust mediation mechanisms and calibrate the reliance on the different functionalities of the same autonomous defense agents. This calibration helps prevent over-reliance on potentially fallible automation while leveraging its strengths, ultimately enhancing the safety and effectiveness of HATs. The assignment of credits and responsibilities is also important in settings such as cybersecurity, where each decision can be of high risk. Cyber workforce includes staff from very different specializations trained in cybersecurity. Special caution is needed when this type of staff is involved in human-autonomous teaming. As they may not have a "fair" judgment on the autonomous agents.

7.2. Limitations and future work

Human-like cognition. We designed an autonomous agent with human cognition based on instance-based learning theory (IBLT), which has enjoyed many successes in replicating individual human-like behavior in the cyber domain and other contexts (Gonzalez, 2024). However, the definition of human likeness in a team context needs further exploration. Future research may focus on conducting "Turing experiments" in a team setting to better understand the desirable teamwork behavior and human-likeness. It is possible that "human-like" is not equivalent to "ideal autonomous teammate", and we might want the autonomous agents to avoid the limitations of humans in teamwork. For example, humans may prefer to reach consensus rather than propose objections and alternative solutions, which has a negative impact on teamwork effectiveness. Secondly, in our current design of the Cognitive agent, the agent leverages the accumulation of instance-based knowledge, recognition-based retrieval, adaptive strategies, and feedback update mechanisms in IBLT. It is unclear how each individual underlying cognitive mechanism contributes to the human-likeness of agents. In future research, we will investigate the instance-based learning mechanisms in more detail to better understand the link between cognitive mechanisms and desirable human-like teamwork behavior. Finally, the current Cognitive agent treats the human teammate as part of the environment, while humans recognize each other's strengths, weaknesses, and working styles to coordinate their actions and adapt their behavior to other team members. In future research, we plan to develop a Cognitive agent that includes a direct observation of human actions, so that we can give the Cognitive agent the ability to make predictions ahead of time regarding human intentions as in the work on Theory of Mid (Nguyen & Gonzalez, 2022). In addition to investigating specific cognitive mechanisms, future work could explore comparisons with other learning approaches, such as reinforcement learning, which has been applied in cybersecurity contexts (Du et al., 2022). While our current study focused on IBL-based cognitive agents, understanding the relative advantages of different learning frameworks for human-autonomy teaming in cybersecurity would provide valuable insights for the field. Specifically, research could examine how different algorithmic approaches affect team dynamics, human trust, and overall performance in various cybersecurity scenarios and team configurations.

Human-automation communication. Our current interactive online team game only supports the action phase of teamwork, where the agent and human participants collectively protect a network through four defensive actions. Teams, in reality, also engage in another important phase of teamwork: the transition phase. It refers to periods of time when teams focus primarily on evaluation and/or planning activities to guide their achievement of a team goal or objective. Teams must take a look at how well they performed during the previous episode and prepare for the next. They compare current performance levels with goals and derive performance gaps. Closing these gaps, in combination with current and anticipated future assignments, guides the development of future performance goals and strategies to achieve them. This phase is especially important in adversarial cyber scenarios, where team members need to develop a threat model and collectively develop a strategy against the attacker. In future work, we will enhance the platform with human–agent communication facilities to support a richer HAT teamwork experience.

HAT for cybersecurity. In the experiment reported in this work, we recruited participants with cybersecurity knowledge from Amazon MTurk. We preselected these participants based on their technical background and screened them according to their cybersecurity knowledge, but they are not necessarily experts in cyber defense. Another limitation comes from the simple cyber defense scenario, which involves a single attacker on a small network. Previous research shows that factors such as workload, stress, and risk can affect teamwork. In cybersecurity, high workload, stress caused by the presence of powerful attackers, and risks involved in defense decisions are very common and cannot be ignored. In future work, we will explore game scenarios with various levels of complexity and workload to better understand the unique challenges of HAT in cybersecurity. Our platform is in continuous development as we refine its capabilities and usability. Motivated by the need to develop open source testbeds and platforms for HAT research, we plan to make our development general and provide documentation for others to consume. The improvements will make it easier for researchers to conduct controlled experiments with minimal coding requirements. We are working to create more accessible interfaces, comprehensive documentation and ready-to-use experimental templates that will lower the entry barrier for researchers of diverse backgrounds. This enhanced platform will allow us to conduct experiments that resemble team configurations in real-world settings, such as including more than two members of the human or autonomous agent team. Such configurations would more accurately reflect operational security teams in which multiple analysts and automated tools work together to defend the network infrastructure.

8. Conclusion

In this work, we found that Cognitive agents are indeed better teammates than Heuristic and Random agents. HATs with Cognitive agents are more effective in protecting the network from breaches, recovering exploited hosts faster, and thus losing fewer points in the team task. Human participants partnered with Cognitive agents also showed higher efficiency. On the other hand, the adaptivity of Cognitive agents renders their behavior relatively unpredictable, which can undermine the trust between humans and agents.

This research makes several important contributions to the field of human-autonomy teaming. First, unlike previous studies that have relied on Wizard of Oz approaches in which humans simulate agent behavior (e.g., Musick et al., 2021), our work uses actual autonomous agents that operate independently according to their programmed cognitive or heuristic mechanisms. This methodological advancement allows us to investigate real human-autonomy interactions rather than perceived ones, providing more ecologically valid insights into the dynamics of HATs.

Second, we introduced the Team Defense Game (TDG), a novel experimental platform specifically designed to study human-autonomy teaming in cyber-defense contexts. The TDG provides researchers with a controlled environment to examine various aspects of HAT collaboration, including conflict resolution, supervision dynamics, and backup behaviors. This platform offers significant advantages over existing



Fig. 9. Distribution of Agent Target Selection Across Steps by Agent Type. The plot shows the proportion of different host types (User, Enterprise, and Op) targeted by Random, Heuristic, and Cognitive agents over 25 steps of the simulation, averaged across all episodes.

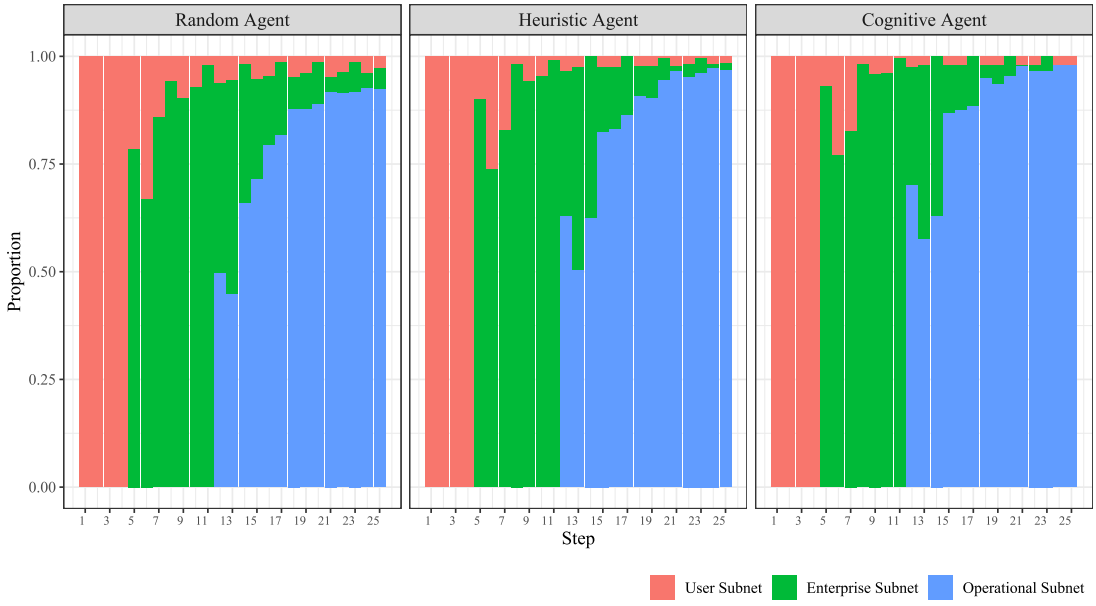


Fig. 10. Temporal Progression of Agent Target Selection by Subnet Type. The plot illustrates the proportion of different subnet types (User, Enterprise, and Operation) targeted by the attacker against Random, Heuristic, and Cognitive agents across 25 simulation steps, averaged across all episodes, demonstrating their strategic priorities throughout the cyber defense scenario.

testbeds by incorporating realistic cyber defense scenarios while maintaining experimental control, making it a valuable resource for future HAT research.

Third, our empirical findings demonstrate that cognitive architectures based on human learning mechanisms offer substantial benefits for HAT effectiveness compared to both random and optimally designed heuristic agents. The cognitive agent's ability to adapt not only to the task environment but also to its human teammate's behavior patterns creates a more complementary partnership, leading to improved team performance despite the potential unpredictability introduced by its adaptive nature.

Using the case of cyber protection teams, this study demonstrated the possibility of effective human-autonomy teamwork and provided evidence of the value of cognitive modeling approaches in agent design. Our results highlight that the implementation of human-like cognitive processes in autonomous agents represents a promising direction for the development of collaborative AI systems. The balance between adaptability and predictability remains a key challenge, but our findings suggest that the benefits of cognitive adaptation outweigh the costs in terms of overall team effectiveness. As autonomous systems become increasingly integrated into human teams in various domains, these insights can inform the design of agents that function not only as tools but as genuine teammates capable of complementing human capabilities and adapting to human work styles.

CRediT authorship contribution statement

Yinuo Du: Writing – original draft, Visualization, Software, Methodology, Data curation, Conceptualization. **Baptiste Prébot:** Writing

– review & editing, Methodology, Data curation, Conceptualization. **Tyler Malloy:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Fei Fang:** Writing – review & editing, Supervision. **Cleotilde Gonzalez:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Ethical approval

This study was approved by the Institutional Review Board of Carnegie Mellon University (approval number STUDY2021_00000324).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Behavior analysis of the autonomous agents

Fig. 9 illustrates the target selection patterns of three different agent types: Random, Heuristic, and Cognitive on the 25 steps of an episode, averaged across all episodes in the experiment.

As evidenced by the consistent distribution across all steps, the Random agent maintains approximately equal proportions of actions targeting the User, Enterprise, and Operational subnets throughout an episode.

In contrast, both the Heuristic and Cognitive agents demonstrate similar dynamic behaviors regarding the action proportions on the three subnets. The Heuristic agent initially executes more actions on the Enterprise subnet, gradually shifting their attention toward the Operational subnet as the episode progresses.

Interestingly, while the Cognitive agent initially demonstrates a similar frequency of actions as those of the Heuristic agent, the Cognitive agent appears to prioritize Operational hosts slightly earlier (around Step 15) and more significantly in later steps than the Heuristic agent. Thus, both the Heuristic and Cognitive agents demonstrate dynamic allocation of their resources to protect the most vulnerable and valuable network assets as the attack progresses. The Cognitive agent demonstrates superior performance given its earlier prioritization of the actions on the Operational subnet.

Importantly, the progression of the adversary through the network topology (shown in Fig. 10) is essentially the same, regardless of the type of agents (Random, Heuristic, or Cognitive) used as a human teammate. The adversary begins in the User subnet, proceeding through the Enterprise subnet, and ultimately targets the critical Operational subnet. These two figures illustrate the competency of the Heuristic and the Cognitive agents to dynamically follow the attacker path of action to block it.

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Update

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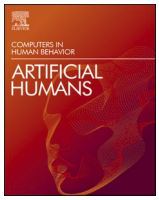
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Erratum regarding corrections related to Ethical approval and informed consent in articles published in [Computers in Human Behavior: Artificial Humans 4C (2025)]

Ethical approval and/or informed consent were not included in the published version of the following articles included in Volume 4C of Computers in Human Behavior: Artificial Humans.

The appropriate Ethical approval and/or informed consent statement, provided by the respective authors, are included below and have been added to the published version of the articles:

1. "What makes children perceive or not perceive minds in generative AI"[CHBAH (2025) 4C: 100135] https://doi.org/10.1016/j.chbah.2025.100135

Ethical approval and informed consent.

This study was approved by the Institutional Review Board of the University of Michigan (approval number HUM00227417). Children were recruited from two public libraries in a mid-west city in the U.S. between July and January 2024. Children participating in our study completed two sessions during story co-creation activities. Signed consent forms were obtained from parents or legal guardians, and verbal assent was obtained for children older than seven years of age. All children consented to participate in the study and were informed that they could discontinue at any time.

2. "A multinational assessment of AI literacy among university students in Germany, the UK, and the US"[CHBAH (2025) 4C: 100132] https://doi.org/10.1016/j.chbah.2025.100132

Ethical approval and informed consent.

This study was approved by the ethical committee of the Technical University of Munich (approval number 2023-251-S-KH). All included participants provided informed consent prior to data collection.

3. "Teaming up with robots: Analysing potential and challenges with healthcare workers and defining teamwork" [CHBAH (2025) 4C: 100136] https://doi.org/10.1016/j.chbah.2025.100136

Ethical approval.

This study was approved by the ethical committee of the German Psychological Society (TransMIT-Zentrum für wissenschaftlich-psychologische Dienstleistungen) (approval number Rosenthal-Von-DerPüttenAstrid2021-04-25VA).

4. "Quid pro Quo: Information disclosure for AI feedback in Human-AI collaboration" [CHBAH (2025) 3C 100137] https://doi.org/10.1016/j.chbah.2025.100137

Ethical approval.

The University of Hamburg uses an ethics questionnaire to assess the need for ethical review board approval. This study was deemed to be exempt from ethical review. The study was conducted ensuring voluntary participation, informed consent, and adherence to good ethical research practices.

5. "Enhancing emotional support in human-robot interaction: Implementing emotion regulation mechanisms in a personal drone" [CHBAH (2025) 4C: 100146] https://doi.org/10.1016/j.chbah.2025.100146

Ethical approval.

This study was approved by the institutional review board of the Ben-Gurion University of the Negev.

6. "Always check important information!" - The role of disclaimers in the perception of AI-generated content" [CHBAH, (2025) 4C: 100142] https://doi.org/10.1016/j.chbah.2025.100142

Informed consent was added to Section 2.1 Method as given below:

2.1. Method.

The study was approved by the ethical committee of the LeibnizInstitut fuer Wissensmedien in Tuebingen (LEK 2021/150). All participants in this study provided informed consent prior to data collection.

7. "Ain't blaming you: Delegation of financial decisions to humans and algorithms" [CHBAH (2025) 4C: 100147] https://doi.org/10.1016/j.chbah.2025.100147

DOIs of original article: https://doi.org/10.1016/j.chbah.2025.100161, https://doi.org/10.1016/j.chbah.2025.100136, https://doi.org/10.1016/j.chbah.2025.100146, https://doi.org/10.1016/j.chbah.2025.100142, https://doi.org/10.1016/j.chbah.2025.100147, https://doi.org/10.1016/j.chbah.2025.100141, https://doi.org/10.1016/j.chbah.2025.100148, https://doi.org/10.1016/j.chbah.2025.100163, https://doi.org/10.1016/j.chbah.2025.100132, https://doi.org/10.1016/j.chbah.2025.100137, https://doi.org/10.1016/j.chbah.2025.100149, https://doi.org/10.1016/j.chbah.2025.100160, https://doi.org/10.1016/j.chbah.2025.100135, https://doi.org/10.1016/j.chbah.2025.100152.

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Ethical Statement.

As per the guidelines of the Ethical Committee of the University of Trento, formal ethics approval was not required as the study involved anonymous adult participants who faced no identifiable risk. The experiment was hosted on Prolific, where participants were informed of the study's nature, duration, and compensation. Participation was voluntary, and consent was obtained by virtue of participants opting to begin the study after reviewing the study specific information that was provided. Participants are free to withdraw at any time.

8. "Evaluating the efficacy of Amanda: A voice-based large language model chatbot for relationship challenges" [CHBAH (2025) 4C: 100141] https://doi.org/10.1016/j.chbah.2025.100141

Informed consent was added in Section 4.2 Participants as given below:

4.2. Participants A total of 309 participants started the study. Of these participants, 49 did not complete the survey, 190 were not eligible due to the potential risk level being too high (e.g., reported suicidal ideation or potential domestic violence) or them not reporting on any specific relationship issue that they wanted to address (i.e., they selected "none" when asked about which issue they would like to address). A further 13 participants were excluded because they either did not interact or complete the interaction with the chatbot or had an issue with the chatbot interaction (their microphone was not working or there was too much noise in the background [e.g., participant was watching TV at the same time]). A further three participants did not complete the follow-up two weeks later (5.3 % dropout rate). Thus, the final sample consisted of 54 participants. Most participants were women (n = 34, 63.0 %), heterosexual (n = 44, 81.5 %), married (n = 27, 50.0 %), white (n = 205, 70.4 %), lived in the UK (n = 29, 53.7 %), and had at least an undergraduate degree (n = 39, 72.2 %). Most participants had also had at least some experience with chatbots (n = 46, 85.2 %) and/or virtual assistants (n = 49, 90.7 %). See Table 1 for the details of demographic variables for each group. All participants that were included in the study provided informed consent before participation.

9. "Socially excluded employees prefer algorithmic evaluation to human assessment: The moderating role of an interdependent culture [CHBAH (2025) 4C: 100152] https://doi.org/10.1016/j.chbah.2025.100152

Ethical approval.

This study was reviewed and approved by the Institutional Review Board of Sophia University (Approval No. 2023-133).

10. "Exploring the connecting potential of AI: Integrating human interpersonal listening and parasocial support into human-computer interactions" [CHBAH (2025) 4C: 100149] https://doi.org/10.1016/j.chbah.2025.100149

Informed consent was added in the 3.1.2. Procedure as given below:

3.1.2. Procedure.

The study received ethical approval from the University of Reading's School of Psychology and Clinical Language Sciences committee (2023-188-NW). All participants were provided with an information sheet and consent form and informed consent from all participants prior to data collection. Following procedures described in Itzchakov et al. (2020), participants were first asked to describe a time when they felt rejected, with the prompt: "Please spend a few moments thinking back to a time when you felt rejected by someone else or other people in your life." Then, participants interacted with Conversational Artificial Intelligence (AI) software (ChatGPT; GPT-3.5) in a back-and-forth conversation. Across six interaction points, the AI assimilated their responses during the conversation and responded as a listener (see coding below) while conveying either high-quality listening or lower-quality listening, based on assignment to condition. Participants responded in text boxes via their online survey, and then reported on the quality of the listening they received.

11. "Experimental evaluation of cognitive agents for collaboration in human-autonomy cyber defense teams" [CHBAH (2025) 4C: 100148] https://doi.org/10.1016/j.chbah.2025.100148

Ethical approval.

This study was approved by the Institutional Review Board of Carnegie Mellon University (approval number STUDY2021_00000324).

12. "Promoting online evaluation skills through educational chatbots" [CHBAH (2025) 4C: 100160] https://doi.org/10.1016/j.chbah.2025.100160

Ethical Declarations.

No formal ethics committee approval was required for this type of study under German regulations and under the policies of the institution where the research was conducted. The study used an anonymous survey and interaction data, no collection of sensitive information, and no interventions that would trigger mandatory ethics review under German university guidelines.

All procedures adhered to the Ethical Principles of Psychologists and Code of Conduct of the American Psychological Association (APA, 2017) and to the ethical guidelines of the Deutsche Gesellschaft für Psychologie and the Berufsverband Deutscher Psychologinnen und Psychologen (DGPs and BDP, 2022). All participants provided informed consent prior to participation.

13. "Cognitive phantoms in large language models through the lens of latent variables" [CHBAH (2025) 4C: 100161] https://doi.org/10.1016/j.chbah.2025.100161

Ethics statement.

This study, procedure and data collection were approved by the local IRB before data collection (reference number TSB_RP1268). Informed consent was provided by all participants prior to commencement of data collection.

14. "Becoming dehumanized by a service robot: An empirical examination of what happens when non-humans perceive us as less than full humans" [CHBAH (2025) 4C: 100163] https://doi.org/10.1016/j.chbah.2025.100163

Ethical approval and informed consent.

This study did not require ethical board review by the Stockholm School of Economics. Participants in this study were recruited through the Prolific platform. Engagement with the platform and study selection is voluntary. Participants were provided with the study specific information prior to data collection and were free to withdraw at any time.

The published version of the following articles included in Volume 5C of Computers in Human Behavior: Artificial Humans contained text that was blinded for peer review and which had not been replaced upon publication.

The appropriate text, provided by the respective authors, are included below and have been added to the published version of the articles:

15. "Knowledge cues to human origins facilitate self-disclosure during interactions with chatbots" [CHBAH, (2025); 5: 100174] https://doi.org/10.1016/j.chbah.2025.100174

Three corrections where [ANONYMIZED FOR THE PEER REVIEW] text has been replaced with the exact text below:

3 Methods

The de-identified data sets, stimuli, and analysis code associated with this study are freely available online at https://osf.io/8vzwp/.

3.3 Ethics.

The research project received ethical approval from the University Ethics Committee (reference number 402210230).

3.4.1 The Chatbots.

Before the interaction with Sam, participants were introduced to it with the following text: "You are going to speak with Sam, a research assistant at the University of Glasgow.

16. "Gaze-informed signatures of trust and collaboration in human-autonomy teams" [CHBAH, 2025; 5: 100171] https://doi.org/10.1016/j.chbah.2025.100171.

The published article contain references to Location 1 and Location 2 text, this has been updated as follows:

2. Material and methods.

2.1. Participants.

A total of 83 volunteers from the University of Colorado, Boulder, and the United States Air Force Academy participated in the experiment. Participants were recruited through newsletter announcements and through the Sona online experimental recruiting software. Three participants were discarded due to technical difficulties with the system and 12 were removed due to poor eye tracking data quality (see Data Processing – Saccades, Fixations and Blinks below and Supplement A.1). As a result, data from 68 participants were used in all analyses. This final sample included 31 participants from the University of Colorado, Boulder (Age: M = 24.1 years, SD = 8; Gender: 20 female, 9 male, 1-non-binary/third gender, 1 preferred not to respond) and 37 participants from the United States Air Force Academy (Age: M = 19.2 years, SD = 1.9; Gender: 20 male, 17 female). Before participating, volunteers signed an informed consent document approved by the institutional review board at the United States Air Force Academy in accordance with the Declaration of Helsinki. Participants at the University of Colorado, Boulder were reimbursed with a $25 gift card while the United States Air Force Academy participants received extra course credit. All participants had normal vision (20/40 or better) without contact lenses. Three participants wore eyeglasses.